Mirror Descent

The sub gradient method: must change prox term

$$x_{t+1} = x_t - \eta g_t, \qquad g_t \in \partial f(x_t)$$

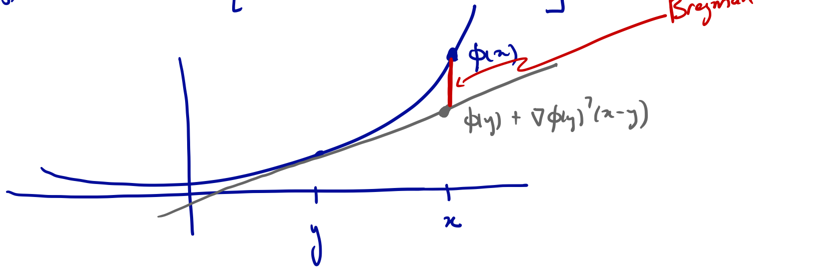
(with set constraint: $x_{t+1} = \mathrm{Proj}_{\mathcal{X}}(x_t - \eta g_t), \quad g_t \in \partial f(x_t)$)

$$x_{t+1} = \underset{x \in \mathcal{X}}{\arg\min} : \eta g_t^T x + \underbrace{\tfrac{1}{2}\|x - x_t\|_2^2}_{\text{will change this.}}$$

Bregman Divergence - definition

For ϕ convex,

$$D_\phi(x,y) \triangleq \phi(x) - \left[\phi(y) + \nabla\phi(y)^T(x-y)\right]$$



Ex: $\phi(x) = \frac{1}{2}\|x\|_2^2$

$$D_\phi(x,y) = \frac{1}{2}\|x\|_2^2 - \left[\frac{1}{2}\|y\|_2^2 + y^T(x-y)\right]$$

$$= \frac{1}{2}\left[\|x\|_2^2 - 2y^Tx + \|y\|_2^2 + \cancel{2\|y\|_2^2}\right]$$

$$= \frac{1}{2}\|x-y\|_2^2$$

Mirror Descent - a first look

$$\boxed{x_{t+1} = \text{argmin}_{x} \; \eta g_t^T x + \frac{1}{2} D_\phi(x, x_t)}$$

If $\phi = \frac{1}{2}\|\cdot\|_2^2$ we recover usual subgradient method.

If $\phi(x) = \frac{1}{2} x^T Q x$, $Q \neq I$, then we get a different alg — (see first lecture in MD series)

Back to Bregman Divergence - properties

$$(\nabla\phi(x) - \nabla\phi(y))^T(x-z) = D_\phi(x,y) + D_\phi(z,x) - D_\phi(z,y)$$

From the definition, RHS:

$$\cancel{\phi(x)} - (\cancel{\phi(y)} + \nabla\phi(y)^T(x-y))$$

$$+ \cancel{\phi(z)} - (\cancel{\phi(x)} + \nabla\phi(x)^T(z-x))$$

$$- \cancel{\phi(z)} + (\cancel{\phi(y)} + \nabla\phi(y)(z-y))$$

$$= \nabla\phi(x)^T(x-z) - \nabla\phi(y)^T(x-y) + \nabla\phi(y)^T(z-y)$$

$$= \nabla\phi(x)^T(x-z) - \nabla\phi(y)^T(x-z)$$

$$= (\nabla\phi(x) - \nabla\phi(y))^T(x-z).$$

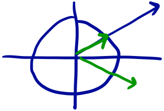
Dual norms - definition

For $\|\cdot\|$ any norm, its <u>dual</u> is defined:

$$\|u\|_* \triangleq \max: \; u^T v \quad \text{st}: \; \|v\| \leq 1$$

Examples: ① $\|\cdot\| = \|\cdot\|_2$

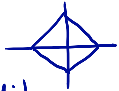
$$(\|u\|_2)_* = \max: \; u^T v \quad \text{st}: \; \|v\| \leq 1$$



maximizer is: $v^* = u/\|u\|$, max value: $u^T v^* = \frac{u^T u}{\|u\|_2} = \|u\|_2$

$\Rightarrow$ Euclidean norm is its own dual (self-dual).

② $\|\cdot\| = \|\cdot\|_1$: $(\|u\|_1)_* = \max: \; u^T v \quad \text{st}: \; \|v\|_1 \leq 1$



$(\|u\|_1)_* = \|u\|_\infty$

$v^* = e_i \cdot \text{sign}(u_i), \quad i = \text{argmax}_i \; |u_i|$

Dual norm - examples

So far: dual of $\|\cdot\|_2$ is $\|\cdot\|_2$ ✓

dual of $\|\cdot\|_1$ is $\|\cdot\|_\infty$ ✓

dual of $\|\cdot\|_\infty$ is $\|\cdot\|_1$ ✓

$$\left(\|u\|_\infty\right)_* = \begin{array}{ll} \max: & u^T v \\ \text{st:} & \|v\|_\infty \le 1 \end{array}$$

$v^* = \text{sign}(u)$ — i.e., $(v^*)_i = \text{sign}(u_i)$

$$u^T v = \sum u_i v_i = \sum u_i \cdot \text{sign}(u_i) = \sum |u_i| = \|u\|_1 .$$

Important Inequality:

$(*)$ $u^T v \leq \|u\|_2 \cdot \|v\|_2$ Cauchy-Schwarz inequality

$u^T v \leq \|u\| \cdot \|v\|_*$ Hölder's inequality

Also: $u^T v \leq \frac{1}{2}\|u\|_2^2 + \frac{1}{2}\|v\|_2^2$ — completing the square.

We will use:

$$u^T v \leq \frac{1}{2\alpha}\|u\|^2 + \frac{\alpha}{2}\|v\|_*^2$$